Expectation-Maximization for RSS noise calibration

A synthetic radio tomographic imaging study

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C++17 implementation and preliminary results

Synthetic observations. No physical Wi-Fi measurements.

RTI and the role of EM

RTI reconstructs attenuation from changes across radio links.

$$y = W x + \text{noise}$$

y: RSS changes on 66 links
W: link-to-pixel sensitivity matrix
x: attenuation in 100 image pixels

EM estimates the noise distribution from centered calibration residuals.

It does not directly output a location.

The target carries no transmitter in the device-free RTI concept.

Questions and experimental design

RQ1 Does parameter error fall as sample size grows?
RQ2 Do different initial values yield similar estimates?
RQ3 How useful is the resulting RTI reconstruction?

Setting	Value
Area and grid	6.4 m square, 10 x 10 pixels
Network	12 perimeter nodes, 66 available links
Main calibration and queries	5,000 residuals, 500 paired location cases
Reproducibility	Seed 7, independent training/query noise

Each query has the same observation vector for all compared methods.

Noise model and hidden component

f(r) = p N(r; 0, s1^2) + (1 - p) N(r; 0, s2^2)

Unknown	Meaning	Generating value
p	Probability of the narrow component	0.548
s1	Narrow-component standard deviation	0.971 dB
s2	Wide-component standard deviation	3.003 dB

Observed: residual r in dB
Hidden: which Gaussian generated that residual

EM pseudocode

```
EM_Noise(r[1..N], Imax, eps)
  initialize theta = (p, s1, s2)
  for t = 1..Imax:
    old = theta
    for i = 1..N:                          // E-step
      a = p*phi(r[i];s1); b = (1-p)*phi(r[i];s2)
      tau[i] = a/(a+b)
    N1 = sum(tau); N2 = N-N1               // M-step
    p = N1/N
    s1 = sqrt(sum(tau[i]*r[i]^2)/N1)
    s2 = sqrt(sum((1-tau[i])*r[i]^2)/N2)
    apply bounds; order components; theta = (p,s1,s2)
    if relative_change(theta,old) < eps: return theta,t,true
  return theta,Imax,false
```

tau is a posterior probability. C++ uses stable log densities and protected divisions. eps = 1e-7, Imax = 500.

One EM iteration by hand

Residuals [-3, -1, 1, 3] dB Initial p = 0.5, s1 = 1, s2 = 3

Residuals	tau: narrow	1 - tau: wide
-1 and +1 dB	0.657949	0.342051
-3 and +3 dB	0.052085	0.947915

New p = (2 x 0.657949 + 2 x 0.052085) / 4 = 0.355017
New s1 = 1.259700 dB New s2 = 2.622727 dB

The larger residuals contribute more weight to the wide component.

Computational cost of EM

N residuals I executed iterations Imax iteration cap
Two fixed components. Basic operation: one log-density evaluation.

$C_E = \Sigma(t=1..I)\ \Sigma(i=1..N)\ \Sigma(j=1..2)\ 1 = 2IN$

Counts sample-component evaluations, not every arithmetic instruction.

Initialization	**Θ(N)**	**Total time: Θ(IN)**
E-step per iteration	**Θ(N)**	**Execution cap: O(Imax N)**
M-step per iteration	**Θ(N)**	**Auxiliary tau vector: Θ(N)**
Parameter check	**Θ(1)**	**Parameters and output: Θ(1)**

Successive E and M scans add. I depends on data, initialization and tolerance.

EM parameter recovery and initialization

N	Converged	MAE p	MAE s1 (dB)	MAE s2 (dB)
100	19/20	0.1048	0.1671	0.4036
500	20/20	0.0375	0.0509	0.1230
1,000	20/20	0.0381	0.0501	0.1140
5,000	20/20	0.0158	0.0194	0.0479
10,000	20/20	0.0109	0.0149	0.0325

20 random starts on one N = 5,000 sample:
20/20 met tolerance in 85-150 iterations, with nearly identical estimates.

Recovery improves broadly with sample size under the model used to generate the data.

Fisher scaling changes regularization

Classical RTI
$\hat{x} = (W^T W + \alpha L)^{-1} W^T y$

Exploratory extension
$\hat{x}_F = (\gamma W^T W + \alpha L)^{-1} \gamma W^T y$

Same RTI form with $\alpha_{effective} = \alpha / \gamma$
$0.10 / 0.41823 = 0.23910$

This comparison alone cannot establish an EM benefit over RTI with tuned regularization.

Localization compared with a center control

Method	Mean error (m)	RMSE (m)
Always predict room center	2.407	2.562
Classical Tikhonov RTI	3.244	3.840
RTI with Fisher scaling	3.062	3.698
K-NN fingerprints	2.618	2.891

Every tested signal-based method has higher mean error than the center control on these 500 queries.

This configuration does not demonstrate useful localization beyond the simple control.

Limits and the next experiment

Current limits

Calibration and test noise share the same model.
One geometry, one main seed and one point target.
No real Wi-Fi measurements or obstacle propagation.

Next experiment

Tune alpha using development data, freeze the choice,
then test on new seeds with the center control included.

A wider application claim requires evidence beyond this synthetic study.

Conclusions

EM recovers known mixture parameters more accurately as calibration sample size grows, on average.

The 20 tested starts agree on one fixed sample.
This result supports local repeatability.

The localization methods fail to beat the center control in mean error. Fisher scaling also changes alpha.

Next step: tune and test the reconstruction before expanding the physical application.